Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: LOCKWOOD III DAIRY

Physical address of dairy:

15343 TIM BELL RD	WATERFORD	Stanislaus	95386
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 03/15/1974

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

XX15-XX15-XX85-XXXX

B. OPERATORS

LEMOS, DARIN DANIEL

Operator name: LEMOS, DARIN DANIEL

Telephone no.: (209) 874-9855 Landline, (209) 614-3608 Cellular

P.O. Box 235	WATERFORD	CA	95386
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

LEMOS, DARIN DANIEL

Legal owner name: LEMOS, DARIN DANIEL

Telephone no.: (209) 874-9855 Landline, (209) 614-3608 Cellular

P.O. Box 235	WATERFORD	CA	95386
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	70	175	195	50	0
Number under roof	495	0	0	0	0	0
Maximum number	495	70	175	195	50	0
Average number	495	70	175	195	50	0
Avg live weight (lbs)	1,400	1,450	1,100	900		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 18,339.33 *tons per reporting period*

Total nitrogen from manure: 224,051.73 *lbs per reporting period*

After ammonia losses (30% loss applied): 156,836.21 *lbs per reporting period*

Total phosphorus from manure: 37,310.82 *lbs per reporting period*

Total potassium from manure: 98,049.22 *lbs per reporting period*

Total salt from manure: 249,167.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,698,000 *gallons*

Total nitrogen generated: 5,876.22 *lbs*

Total phosphorus generated: 6,517.01 *lbs*

Total potassium generated: 70,988.65 *lbs*

Total salt generated: 197,601.26 *lbs*

	4,698,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,698,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
AG IRRIGATION WELL	Ground water
MID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
07/25/2023	Corral solids	2,016.00 *ton*	Dry-weight	24.7		15,380.00	11,005.00	30,495.00		50.20
10/26/2023	Corral solids	1,800.00 *ton*	Dry-weight	25.2		32,500.00	18,920.00	59,160.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	134,211.16	84,360.01	251,891.80	1,524,120.19
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	134,211.16	84,360.01	251,891.80	1,524,120.19

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
FIELD 1	31	31	2	process wastewater	XX15-XX15-XX85-XXXX
FIELD 2	35	35	2	process wastewater	XX15-XX15-XX85-XXXX
PASTURE	5	5	0	none	XX15-XX15-XX85-XXXX
Totals for areas that were used for application	66	66	4		
Totals for areas that were not used for application	5	5	0		
Land application area totals	71	71	4		

B. CROPS AND HARVESTS

FIELD 1

Field name: FIELD 1

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	455.70 *ton*	Dry-weight		73.0	22,200.00	3,700.00	36,000.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.70	176.22	29.37	285.77	912.87

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,056.80 *ton*	Dry-weight		73.4	3,800.00	700.00	4,700.00		1.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.09	68.92	12.70	85.24	326.45

FIELD 2

Field name: FIELD 2

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	514.50 *ton*	Dry-weight		72.4	23,600.00	4,000.00	38,700.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.70	191.50	32.46	314.03	876.36

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,193.20 *ton*	Dry-weight		74.6	3,900.00	800.00	4,800.00		2.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.09	67.54	13.85	83.13	381.01

NUTRIENT BUDGET

A. LAND APPLICATIONS

FIELD 1 - 11/08/2022: Oats, silage-soft dough

Field name: FIELD 1

Crop: Oats, silage-soft dough — Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	43.61	4.88	139.55	566.92	324,000.00 *gal*
MID Canal	Surface water	0.89	0.00	0.00	0.00	3,300,000.00 *gal*
Application event totals		44.50	4.88	139.55	566.92	

FIELD 1 - 06/23/2023: Corn, silage

Field name: FIELD 1

Crop: Corn, silage — Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.92	0.00	0.00	0.00	3,400,000.00 *gal*
Application event totals		0.92	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.92	0.00	0.00	0.00	3,400,000.00 *gal*
Application event totals		0.92	0.00	0.00	0.00	

FIELD 1 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	59.60	5.09	54.10	457.90	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	2,850,000.00 *gal*
Application event totals		60.37	5.09	54.10	457.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	3.05	13.95	139.77	341.61	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	2,850,000.00 *gal*
Application event totals		3.82	13.95	139.77	341.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	3.05	13.95	139.77	341.61	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	2,850,000.00 *gal*
Application event totals		3.82	13.95	139.77	341.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.44	11.16	111.81	273.29	216,000.00 *gal*
MID Canal	Surface water	0.71	0.00	0.00	0.00	2,650,000.00 *gal*
Application event totals		3.16	11.16	111.81	273.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.44	11.16	111.81	273.29	216,000.00 *gal*
MID Canal	Surface water	0.71	0.00	0.00	0.00	2,650,000.00 *gal*
Application event totals		3.16	11.16	111.81	273.29	

FIELD 1 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.44	11.16	111.81	273.29	216,000.00 *gal*
MID Canal	Surface water	0.71	0.00	0.00	0.00	2,650,000.00 *gal*
Application event totals		3.16	11.16	111.81	273.29	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.44	11.16	111.81	273.29	216,000.00 *gal*
MID Canal	Surface water	0.71	0.00	0.00	0.00	2,650,000.00 *gal*
Application event totals		3.16	11.16	111.81	273.29	

09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.77	0.00	0.00	0.00	2,850,000.00 *gal*
Application event totals		0.77	0.00	0.00	0.00	

FIELD 2 - 11/08/2022: Oats, silage-soft dough

Field name: FIELD 2

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	3.24	14.83	148.55	363.08	324,000.00 *gal*
MID Canal	Surface water	0.92	0.00	0.00	0.00	3,850,000.00 *gal*
Application event totals		4.16	14.83	148.55	363.08	

FIELD 2 - 06/23/2023: Corn, silage

FIELD 2 - 06/23/2023: Corn, silage

Field name: FIELD 2

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.92	0.00	0.00	0.00	3,850,000.00 *gal*
Application event totals		0.92	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	3.24	14.83	148.55	363.08	324,000.00 *gal*
MID Canal	Surface water	0.92	0.00	0.00	0.00	3,850,000.00 *gal*
Application event totals		4.16	14.83	148.55	363.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	3.24	14.83	148.55	363.08	324,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	3,220,000.00 *gal*
Application event totals		4.01	14.83	148.55	363.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	38.63	4.33	123.60	502.13	324,000.00 *gal*
Application event totals		38.63	4.33	123.60	502.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	3.24	14.83	148.55	363.08	324,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	3,220,000.00 *gal*
Application event totals		4.01	14.83	148.55	363.08	

FIELD 2 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.70	12.36	123.79	302.57	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	3,220,000.00 *gal*
Application event totals		3.47	12.36	123.79	302.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.70	12.36	123.79	302.57	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	3,220,000.00 *gal*
Application event totals		3.47	12.36	123.79	302.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.70	12.36	123.79	302.57	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	3,220,000.00 *gal*
Application event totals		3.47	12.36	123.79	302.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	2.70	12.36	123.79	302.57	270,000.00 *gal*
MID Canal	Surface water	0.77	0.00	0.00	0.00	3,220,000.00 *gal*
Application event totals		3.47	12.36	123.79	302.57	

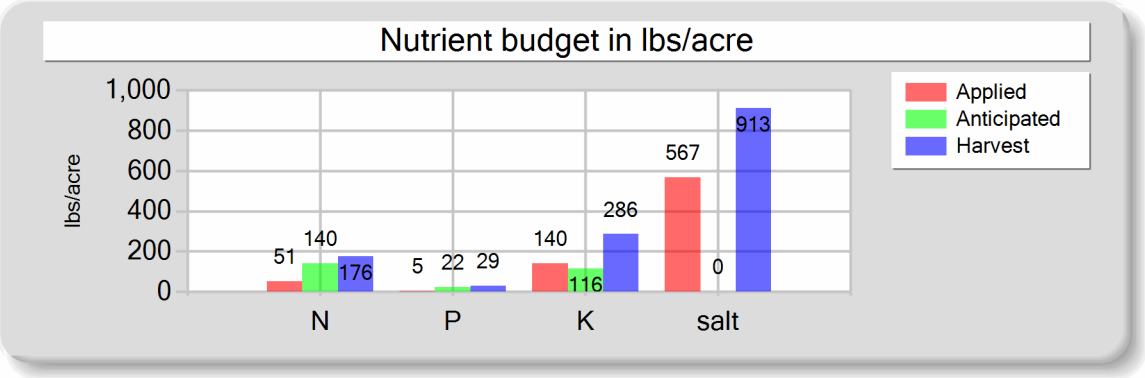
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.92	0.00	0.00	0.00	3,850,000.00 *gal*
Application event totals		0.92	0.00	0.00	0.00	

B. NUTRIENT BUDGET

FIELD 1 - 11/08/2022: Oats, silage-soft dough

Field name: FIELD 1 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	43.61	4.88	139.55	566.92
Fresh water	0.89	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	51.50	4.88	139.55	566.92
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	176.22	29.37	285.77	912.87
Nutrient balance	-124.73	-24.49	-146.22	-345.95
Applied to removed ratio	0.29	0.17	0.49	0.62

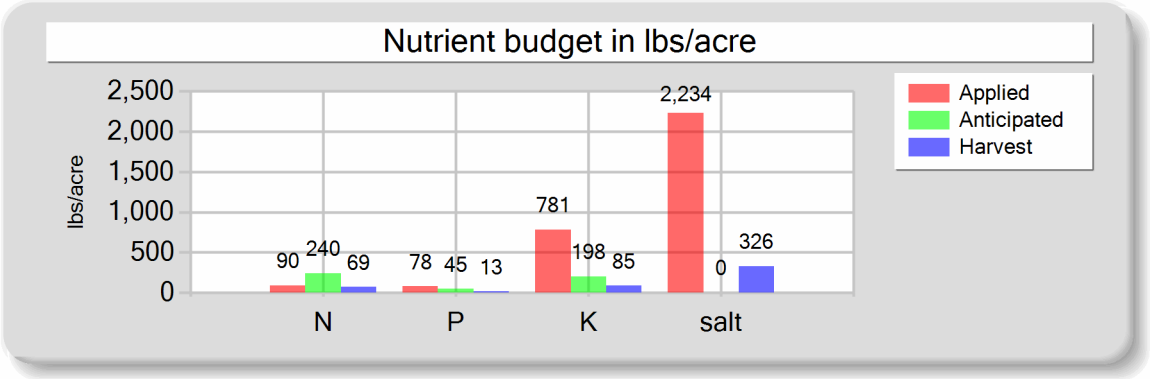
Fresh water applied	
3,300,000.00	*gallons*
121.53	*acre-inches*
3.92	*inches/acre*

Process wastewater applied	
324,000.00	*gallons*
11.93	*acre-inches*
0.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

FIELD 1 - 06/23/2023: Corn, silage

Field name: FIELD 1 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	75.47	77.65	780.90	2,234.25
Fresh water	7.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	90.23	77.65	780.90	2,234.25
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	68.92	12.70	85.24	326.45
Nutrient balance	21.31	64.96	695.66	1,907.80
Applied to removed ratio	1.31	6.12	9.16	6.84

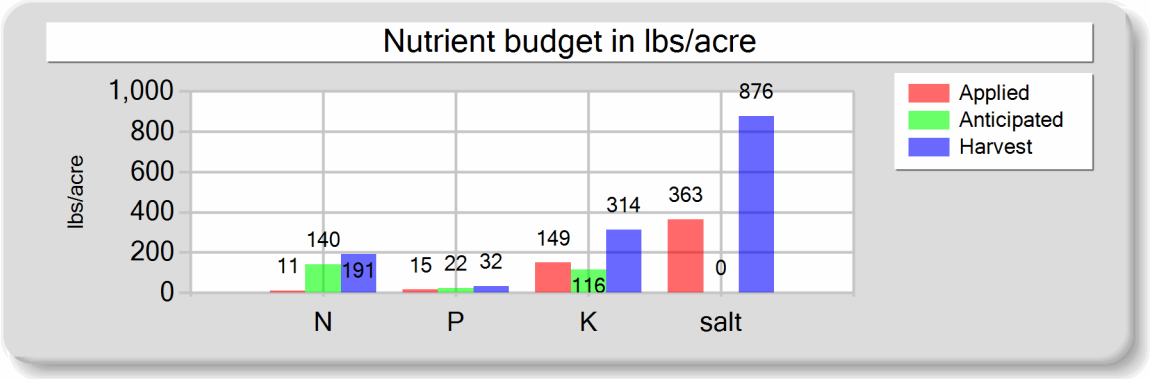
Fresh water applied	
28,800,000.00	*gallons*
1,060.61	*acre-inches*
34.21	*inches/acre*

Process wastewater applied	
1,674,000.00	*gallons*
61.65	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

FIELD 2 - 11/08/2022: Oats, silage-soft dough

Field name: FIELD 2 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	3.24	14.83	148.55	363.08
Fresh water	0.92	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	11.16	14.83	148.55	363.08
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	191.50	32.46	314.03	876.36
Nutrient balance	-180.34	-17.63	-165.47	-513.28
Applied to removed ratio	0.06	0.46	0.47	0.41

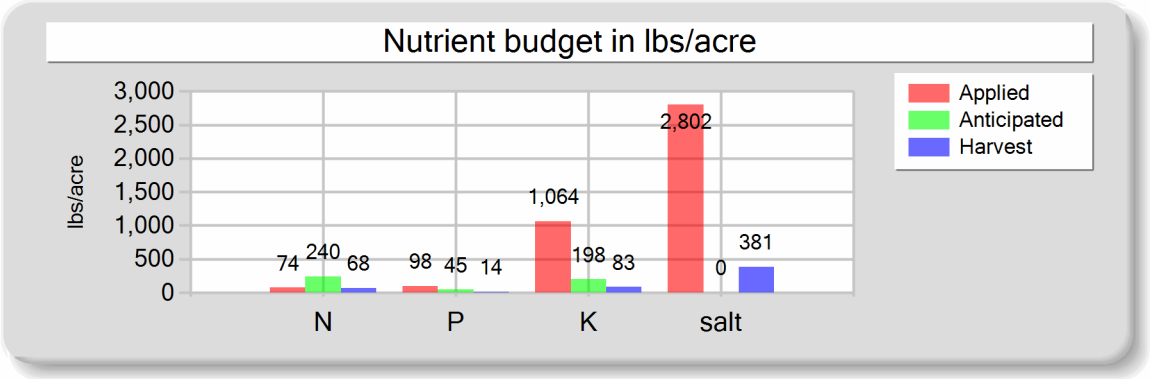
Fresh water applied	
3,850,000.00	*gallons*
141.78	*acre-inches*
4.05	*inches/acre*

Process wastewater applied	
324,000.00	*gallons*
11.93	*acre-inches*
0.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

FIELD 2 - 06/23/2023: Corn, silage

Field name: FIELD 2 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	59.17	98.26	1,064.44	2,801.63
Fresh water	7.36	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	73.53	98.26	1,064.44	2,801.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	67.54	13.85	83.13	381.01
Nutrient balance	5.99	84.41	981.31	2,420.63
Applied to removed ratio	1.09	7.09	12.80	7.35

Fresh water applied	
30,870,000.00	*gallons*
1,136.84	*acre-inches*
32.48	*inches/acre*

Process wastewater applied	
2,376,000.00	*gallons*
87.50	*acre-inches*
2.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

579461-01

Sample and source description: 579461-01
Sample date: 09/29/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 25.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,500.00	18,920.00	59,160.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

579462-01

Sample and source description: 579462-01
Sample date: 09/30/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 24.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,380.00	11,005.00	30,495.00	8,350.00	3,850.00	2,900.00	0.00	0.00		50.20
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

3RD QUARTER LAGOON

Sample and source description: 3RD QUARTER LAGOON
Sample date: 09/28/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	500.00	460.00	0.00	0.00	56.00	1,600.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10,000.00	6,500
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

L2.2-14R64280

Sample and source description: L2.2-14R64280

Sample date: 06/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	42.00	0.00	0.00	0.00	192.00	1,923.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8,400.00	4,700
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

L2.2-114R66432

Sample and source description: L2.2-114R66432

Sample date: 12/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	800.00	470.00	0.00	20.00	70.00	744.40	100.00	50.00	50.00	0.00	0.00	50.00	100.00	8,800.00	6,300
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

C. FRESH WATER ANALYSES

AG IRRIGATION WELL

L2.2-14R66305

Sample description: L2.2-14R66305

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	110.00	0.00	110.00	88.00	28.00	100.00	297.00	0.00	69.00	87.00	1,100.00	790
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

MID Canal

MID Canal

CANAL SAMPLE

Sample description: CANAL SAMPLE

Sample date: 10/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

FIELD 1 - 11/08/2022: Oats, silage-soft dough

60059901

Sample and source description: 60059901

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	3,700.00	36,000.00		11.50
DL	1.00	1.00	1.00		1.00

FIELD 1 - 06/23/2023: Corn, silage

FIELD 1 - 06/23/2023: Corn, silage

61527401

Sample and source description: 61527401

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	4,700.00		1.80
DL	1.00	1.00	1.00		1.00

FIELD 2 - 11/08/2022: Oats, silage-soft dough

60059902

Sample and source description: 60059902

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	4,000.00	38,700.00		10.80
DL	1.00	1.00	1.00		1.00

FIELD 2 - 06/23/2023: Corn, silage

61527402

Sample and source description: 61527402

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	800.00	4,800.00		2.20
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

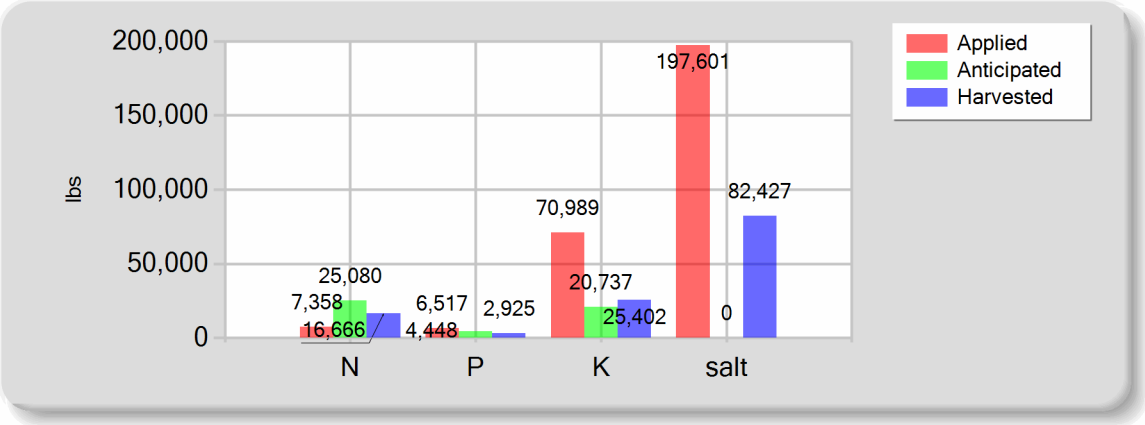
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

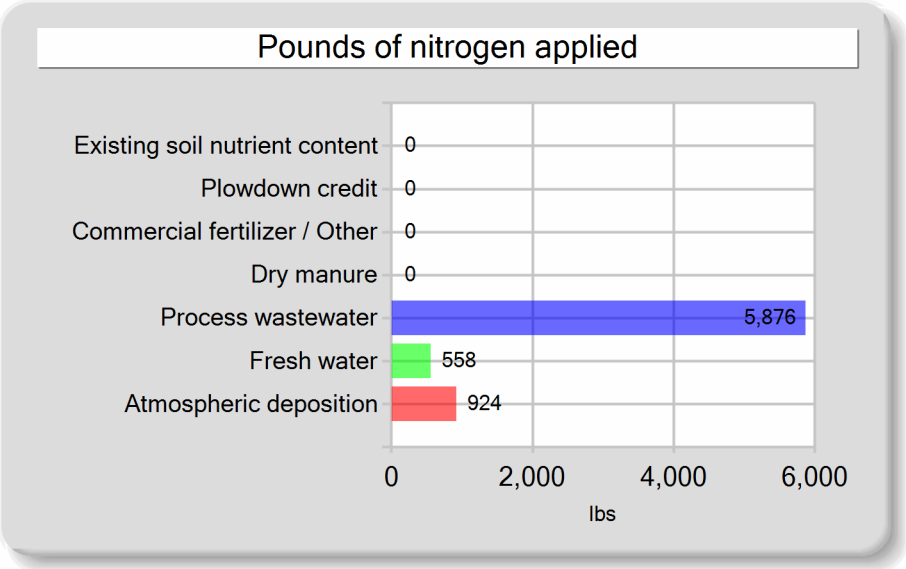
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

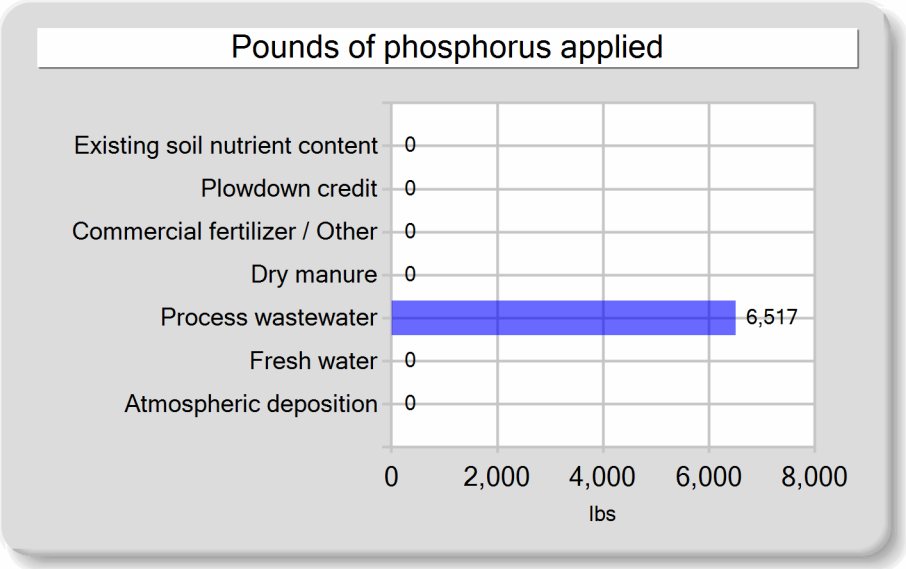
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	5,876.22	6,517.01	70,988.65	197,601.26
Fresh water	557.61	0.00	0.00	0.00
Atmospheric deposition	924.00	0.00	0.00	0.00
Total nutrients applied	7,357.83	6,517.01	70,988.65	197,601.26
Anticipated crop nutrient removal	25,080.00	4,448.40	20,737.20	0.00
Actual crop nutrient removal	16,665.82	2,924.97	25,401.68	82,426.52
Nutrient balance	-9,307.99	3,592.04	45,586.96	115,174.73
Applied to removed ratio	0.44	2.23	2.79	2.40

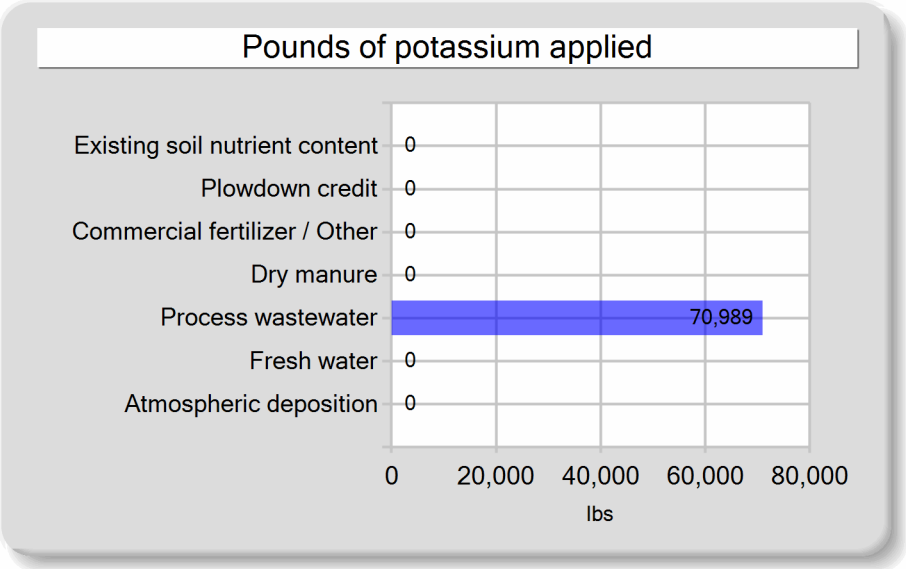
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

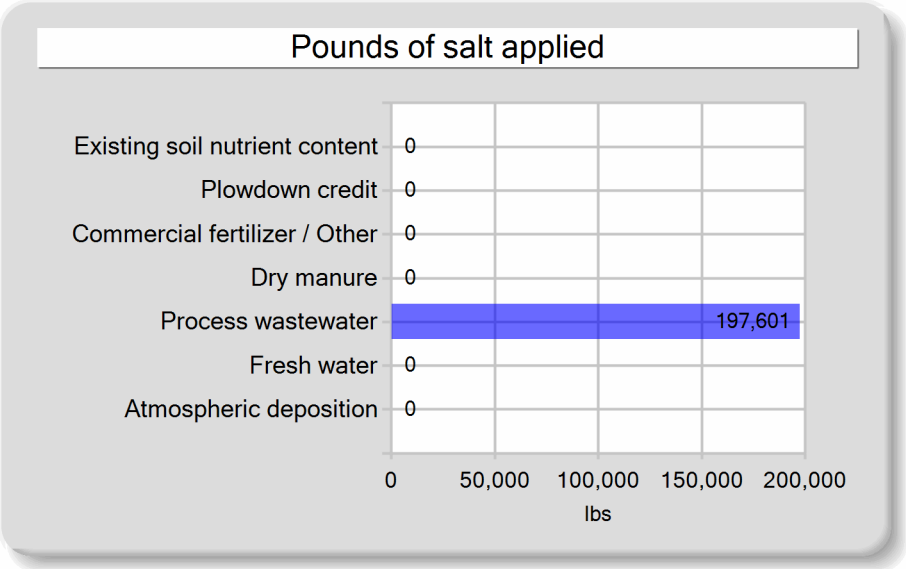


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

The previous consultant was not communicating well with Mr. Lemos. Mr. Lemos was not aware that samples were not being taken. He has started with a new consultant as of 5/24.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Darin Lemos	SAME
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
DARIN DANIEL LEMOS	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/20/24	6/20/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Darin Lemos

Name of Dairy Facility: Lockwood III Dairy

Facility Address: 15343 Timbell rd (Number and Street) Waterford (City) CA (Zip Code) 95386

Contact Person Name and Phone Number: Darin Lemos (Name) 209-614-3608 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: 5M Hay + Ag

Address of Hauling Company /Person: 1831 Hazeldem Rd (Number and Street) Waterford (City), Ca (Zip Code) 95386

Contact Person: Victor Mendoza (Name) 209-602-1672 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

5M Hay & Ag (Name) 21631 E. Keyes Rd (Number and Street) Montpelier (City) C (Zip Code) 95316 209602-1672 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

(Number and Street) SAME (City) (Zip Code) 020006027 (Assessor's Parcel Number)

Dates Hauled: 7/20/23 thru 7/25/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 3200 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): 1260 lbs/cu-

Method used to determine amount of manure: 62.5 loads 40 yard per load
Side dump trailers 20yds each trailer Set of doubles

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes ___X___ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Martin Lemos Date: 10/28/23

Hauler's Signature: [illegible] Date: 10/28/23

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Darin Lemos

Name of Dairy Facility: Lockwood III Dairy

Facility Address: 15343 Tim Bell rd. Waterford CA 95386
Number and Street City Zip Code

Contact Person Name and Phone Number: Darin Lemos 209-634-3608
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: SM Hay & Ag

Address of Hauling Company /Person: 1831 Hazeldean rd. Waterford CA 95386
Number and Street City Zip Code

Contact Person: Victor Mendoza 209-602-1672
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

SM Hay & Ag 21631 E. Keyes Rd. Montpelier CA 95316 209-602-1672
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

SAME 020006027
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 10/25/23 thru 10/26/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2500 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _____
Manure Density (if amount reported in cubic yards): 1440 lbs/cu

Method used to determine amount of manure: 80 loads 40yards per load Side dump trailers 20yd each set of doubles

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes ✓ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Darin Lemos Date: 9-10-2023

Hauler's Signature: [signature] Date: 9-10-2023